CLAIMS

What is claimed is:

1. A control device comprising:

a passive infrared sensing circuit comprising:

a pyroelectric detector configured to receive infrared energy and generate an output signal in response to the received infrared energy, the pyroelectric detector configured to generate the output signal as an alternating-current signal having a direct-current offset during a operational state of the passive infrared sensing circuit; and

a first amplifier circuit comprising a first operational amplifier having an inverting input, a non-inverting input, and an output, the non-inverting input coupled to receive the output signal of the pyroelectric detector, the first amplifier circuit further comprising a series combination of a first resistor and a first capacitor coupled between the inverting input of the first operational amplifier and the circuit common, and a parallel combination of a second resistor and a second capacitor coupled between the inverting input and the output of the first operational amplifier, the first amplifier circuit further comprising a diode having an anode coupled to the output of the first operational amplifier and a cathode coupled to the inverting input of the first operational amplifier, the first operational amplifier configured to amplify the output signal of the pyroelectric detector to generate the amplified signal at the output of the first operational amplifier during the operational state of the passive infrared sensing circuit; and

a control circuit coupled to receive a passive infrared sensing signal from the passive infrared sensing circuit;

wherein, prior to the passive infrared sensing circuit entering the operational state, the first operational amplifier is configured to drive a voltage at the output high towards a supply voltage and the diode is configured to conduct at least a portion of a first charging current from the supply voltage to charge the first capacitor of the first amplifier circuit, the first capacitor configured to

charge to a magnitude that is approximately equal to the direct-current offset of the output signal of the pyroelectric detector.

2. The control device of claim 1, further comprising:

a second amplifier circuit comprising a second operational amplifier having an inverting input, a non-inverting input, and an output, the non-inverting input coupled to receive the amplified signal, the second amplifier circuit further comprising a series combination of a third resistor and a third capacitor coupled between the inverting input of the second operational amplifier and the circuit common, and a parallel combination of a fourth resistor and a fourth capacitor coupled between the inverting input and the output of the second operational amplifier.

3. The control device of claim 2, wherein the control circuit is further coupled to the junction of the third resistor and the third capacitor of the second operational amplifier, the control circuit configured to, prior to the passive infrared sensing circuit entering an operational state, charge the third capacitor of the second amplifier circuit by conducting a second charging current through the control circuit and the third capacitor, and stop charging the third capacitor when the magnitude of the sampled voltage is greater than or equal to a threshold voltage.

4. The control device of claim 3, wherein the control circuit comprises a processor having a port configured to be driven high for charging the third capacitor and conducting the second charging current through the processor.

5. The control device of claim 4, wherein the passive infrared sensing circuit further comprises a reference voltage circuit configured to generate a reference voltage, the reference voltage circuit having a fifth capacitor across which the reference voltage is developed, the voltage reference circuit configured to couple the reference voltage to the amplified signal at the non-inverting input of the operational amplifier of the second amplifier circuit, such that the amplified signal has a direct-current offset equal to approximately the reference voltage.

6. The control device of claim 5, wherein the processor is configured to configure to simultaneously charge the third capacitor and the fifth capacitor so that the magnitudes of the voltages across the third and fifth capacitors increase at approximately the same rate.

7. The control device of claim 4, wherein the port of the processor is coupled to the junction of the first resistor and the first capacitor via a fifth resistor, the processor configured to charge the third capacitor by driving a voltage at the port high towards a supply voltage and conducting the second charging current through the fifth resistor.

8. The control device of claim 7, wherein the processor is configured to periodically configure the port as an input, sample the voltage at the port, and configure the port as an output prior to driving the voltage at the port high to charge the third capacitor, the processor further configured to stop charging the first capacitor when the magnitude of the sampled voltage is greater than or equal to a threshold voltage.

9. The control device of claim 4, wherein the port of the processor is coupled to the junction of the third resistor and the third capacitor via a charging circuit, the processor configured to charge the third capacitor by driving a voltage at the port high towards a supply voltage and conducting the second charging current through the charging circuit, the charging circuit comprising a comparator configured to control the current source to stop charging the third capacitor when the magnitude of a capacitor voltage developed across the third capacitor exceeds a threshold voltage.

10. The control device of claim 3, wherein the processor is configured to receive the passive infrared sensing signal from the second amplifier circuit of the passive infrared sensing circuit, the processor configured to detect an occupancy condition or a vacancy condition in response to the passive infrared sensing signal.

11. The control device of claim 10, further comprising:

a controllably conductive device adapted to be coupled between a power source and an electrical load for controlling an amount of power delivered to the electrical load;

wherein the processor is configured to control the controllably conductive device to adjust the amount of current delivered to the electrical load in response to detecting the occupancy condition or the vacancy condition via the passive infrared sensing circuit.

12. The control device of claim 10, further comprising:

a communication circuit configured to transmit messages;

wherein the processor is configured to cause the communication circuit to transmit a message indicating the occupancy condition or the vacancy condition in response to detecting the occupancy condition or the vacancy condition via the passive infrared sensing circuit.

13. The control device of claim 1, wherein a gain of the first amplifier circuit is greater than +/-50.

14. The control device of claim 13, wherein, during the operational state of the passive infrared sensing circuit, a magnitude of a voltage across the parallel combination of the second resistor and the second capacitor does not exceed a forward voltage of the diode.

15. The control device of claim 1, wherein a resistance of the second resistor of the first amplifier circuit is greater than 1 MΩ.

16. A passive infrared sensing circuit comprising:

a pyroelectric detector configured to receive infrared energy and generate an output signal in response to the received infrared energy, the pyroelectric detector configured to generate the output signal as an alternating-current signal having a direct-current offset during an operational state of the passive infrared sensing circuit;

an operational amplifier having an inverting input, a non-inverting input, and an output, the operational amplifier configured to receive power from a supply voltage that is referenced to a circuit common, the non-inverting input of the operational amplifier coupled to receive the output signal of the pyroelectric detector;

a series combination of a first resistor and a first capacitor coupled between the inverting input of the operational amplifier and the circuit common;

a parallel combination of a second resistor and a second capacitor coupled between the inverting input and the output of the operational amplifier; and

a diode having an anode coupled to the output of the operational amplifier and a cathode coupled to the inverting input of the operational amplifier;

wherein, prior to the passive infrared sensing circuit entering the operational state, the operational amplifier is configured to drive a voltage at the output high towards the supply voltage and the diode is configured to conduct at least a portion of a charging current from the supply voltage to charge the first capacitor, the first capacitor configured to charge to a magnitude that is approximately equal to the direct-current offset of the output signal of the pyroelectric detector.

17. The passive infrared sensing circuit of claim 16, wherein, during the operational state of the passive infrared sensing circuit, the operational amplifier is configured to amplify the output signal of the pyroelectric detector to generate an amplified signal at the output of the operational amplifier.

18. The passive infrared sensing circuit of claim 17, wherein the passive infrared sensing circuit comprises an amplifier circuit including the operational amplifier, the first resistor, the first capacitor, the second resistor, the second capacitor, and the diode; and

wherein a gain of the amplifier circuit is greater than +/-50.

19. The passive infrared sensing circuit of claim 18, wherein, during the operational state of the passive infrared sensing circuit, a magnitude of a voltage across the parallel combination of the second resistor and the second capacitor does not exceed a forward voltage of the diode.

20. The passive infrared sensing circuit of claim 17, wherein a resistance of the second resistor is greater than 1 MΩ.